Chemical commodity price forecast based on multi factor combination model

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Abstract

Chemical products are the cornerstone of agricultural development and provide important raw materials for industrial and agricultural production. Their quality, quantity and price stability are closely watched by upstream and downstream industries. Therefore, it is of great practical and theoretical significance to accurately predict the prices of various chemical products. Due to the influence of various factors on the price of chemical products, they show strong nonlinear, non-stationary and no obvious trend characteristics. It is difficult for the traditional single model to capture the internal hidden laws of their data. Based on this, this paper proposes a combination forecasting model based on data decomposition, which can deeply mine the potential volatility characteristics within the data, so as to better grasp the volatility law and realize its price forecast. Guided by the idea of "decomposing input and combining output", this paper constructs a price forecasting model based on multi factor decomposition and integration, and explains and forecasts it from the factor level.

Keywords- EEMD; decomposition; integration; DTW; price- forecast; LSTM

1. Introduction

In recent years, the methods used by scholars to predict the price of bulk commodities can be divided into the following categories: univariate model, multivariate model, univariate multi-scale model, and integrated portfolio model. In this paper, we mainly consider the artificial intelligence model with adaptive ability. Compared with the traditional econometric model, it has natural advantages. It mainly depends on the powerful computing ability of the computer to carry out adaptive learning iteration to adjust parameters in order to achieve the purpose of price prediction.

This paper focuses on the use of short - and long-term neural networks (LSTM) to forecast the price series volatility based on multiple factors. It can be divided into the following two steps: (1) ADF test is conducted on the stability of the price series and the possible influencing factor series, and then Granger causality test is used to screen out the influencing factors that play a role in price forecasting. (2) EEMD is applied to decompose the original sequence into several meaningful IMF, whose main purpose is to process and denoise the nonlinearity of the price sequence. (3) The original sequence after decomposition is reconstructed by DTW algorithm to reduce the possibility of mode aliasing and error accumulation.

Based on the above, relevant factors are substituted into the prediction model through factor screening, and the LSTM algorithm is further combined to build a product price prediction model to mine the internal and external characteristics of the data and improve the prediction accuracy. This chapter focuses on the price prediction of ethylene glycol chemical products. In order to avoid the accidental effect of the model, the model is also used to predict the price of p-xylene.

2. Overview of Factors Affecting the Price of Chemical Products

The price sequence of bulk chemical products is volatile and complex, and will also be affected by many factors, such as the dollar index, the import and export tax rate of chemical products, the value-added tax rate, the Chinese chemical price index CCPI, and relevant national policies. Faced with a large group of factors.

It is impossible to substitute all factors into the model, which will not only bring huge computation, but also reduce the accuracy of the model due to the interference of many factors. Therefore, we must conduct factor screening before real

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modeling.

The factors affecting the price of chemical products can be considered from two aspects: first, like other ordinary commodities, they will be affected by the market, such as the opposition between supply and demand, and the competition between substitutes. Secondly, the price of chemical products is greatly affected by its upstream energy, international petroleum. As an important international energy, petroleum will be affected by the international situation, such as political turbulence and international conflicts, which will indirectly affect the price of chemical products through the price of petroleum. For example, the conflict between Russia and Ukraine in 2022 caused a sudden change in the global situation, and the international crude oil soared during the Asian trading period, thus driving the turbulence of the chemical market and changing the fluctuation trend of chemical product prices. The possible factors affecting the price of chemical products are shown in Table 1.

Table 1: Chemical products price impact factors.

category	influence factor
market	Supply and demand environment
market	Oil price
Economics	China Chemical Price Index (CCPI)
Economics	Exchange rate of RMB against USD (middle rate)
Economics	USD index (right axis)
Economics	S&P500 (right axis)
Economics	Spot gold price (Right Axis)
Economics	CRB Commodity Price Index (Right Axis)
Economics	VAT rate (imported liquefied petroleum gas, natural gas and coal)
event	financial crisis
event	international war

It can be seen from the table that the factors affecting the price of chemical products can be roughly divided into three categories: market factors, economic factors and event factors. Considering the availability of data, this paper focuses on economic factors.

3. Introduction to Related Algorithms

3.1 Data decomposition method EEMD

EEMD was proposed by Huang[1] et al. It is an improved algorithm based on EMD. Since EMD algorithm does not require too much prior knowledge, and can effectively mine the potential information in the data, capture the hidden patterns in the sequence, it is a strong self-adaptability, strong completeness decomposition method, so it has been applied to the data decomposition work by the majority of scholars since it was proposed[2].

However, in the process of using EMD algorithm, its shortcomings of mode aliasing have been gradually discovered, and gradually can't meet the requirements of scholars for data decomposition. Therefore, Huang and Wu[3] et al. have improved on the basis of EMD by adding white noise to the original sequence and using EMD decomposition for many times to finally average multiple sequences, that is, EEMD algorithm to try to prevent the generation of mode aliasing.

Here, the EMD algorithm is briefly introduced. This method uses the Hilbert Huang transform (HHT) to transform the original data x_T is decomposed into several eigenmode functions (IMFs) and a residual sequence (Res), as shown in equation (1) below:

$$x_t = \sum_{i=1}^{N-1} IMF_t(i) + r_t \qquad (1)$$

Where x_t represents the original sequence, $IMF_t(i)$ represents the ith eigenmode function corresponding to the original sequence, r_t is the residual sequence, which can be expressed as the long-term trend corresponding to the sequence.

EEMD algorithm is an improvement of EMD, which is to add white noise to the original sequence many times based on EMD, and then use EMD to decompose.

After averaging the obtained components, the components of each order corresponding to the final EEMD can be obtained.

3.2 DTW reconstruction algorithm

DTW algorithm is based on the idea of dynamic programming to calculate the distance (also known as similarity) between Q sequence with length m and C sequence with length n, where Q =q_1, q_2,…, q_i,…, q_n; C = c_1, c_2,…, c_j,…, c_m;

If m=n, you can directly calculate the distance between two sequences. If m ≠ n, you need to construct a matrix of n x m.

Matrix elements (i, j) represent q_i and c_j The distance d (q_i,c_j) between two points (the distance between each point of sequence Q and each point of C) is calculated by Euclidean distance: d (q_i,c_j) = $(q_i - c_j)^2$. (i, j) indicates point q_i and c_j alignment. Dynamic programming algorithm can be expressed as finding a route to reach the other side through a few points in the grid and ensuring that the route has the shortest distance among all possible routes. Here P = P_1, P_2 … P_k represents the regular path length between two sequences, where k is the path serial number. The expression of the shortest path is shown in equation (2).

$$\boldsymbol{L_{DTW}}(\mathbf{Q},\mathbf{C}) = \boldsymbol{min_p} \sum_{k=1}^{K} \boldsymbol{d}(\boldsymbol{p_k}) \qquad (2)$$

Where L_DTW (Q，C) represents the shortest distance between sequence Q and sequence C. That is, the degree of similarity between two sequences. The DTW reconstruction rule is to cluster two (or more) modes that are close to each other. In particular, because the first mode component has the highest complexity among all modes, and there is no previous mode for distance comparison, the DTW reconstruction rule usually models the IMF1 of the decomposed sequence separately[4].

3.3 LSTM model

LSTM network model is a further improvement of recurrent neural network (RNN), which was first proposed by Sepp Hochreiter and Jiirgen Schmidhuber in 1997[5]. Cyclic neural networks can usually be used to process sequential data. Compared with general neural networks, recurrent neural networks can remember the relationship between data in a sequence. But when the amount of data is too large, because the RNN network has no forgetting mechanism, it will substitute all the previous data into the network without filtering to participate in the calculation, which not only has a huge amount of calculation, but also affects the final prediction results. At the same time, for the phenomenon of gradient disappearance and gradient explosion that may occur in the training process of a long sequence, the important parameters in the network will stop updating or jump updating, and the model will not be able to learn to find the optimal parameters. Considering the above problems of RNN, LSTM network adds a forgetting gate mechanism to the original RNN network. Compared with RNN, LSTM can perform better in long sequence data training. LSTM network structure is shown in Figure 1:

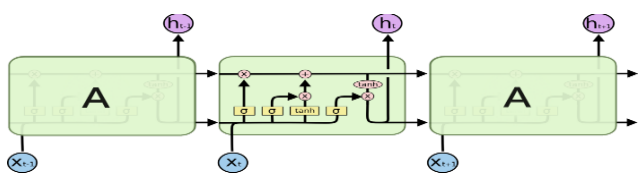

Figure 1: LSTM Network Structure.

LSTM mainly filters information through various gate mechanisms and gives different weights. The main calculation process is as follows:

Step1: Filtering of discarded information

The first step of LSTM is to filter information. This process is completed through the forgetting gate. The amount of information retained is determined by the weight parameter, as shown in the following formula:

$$f_t = \sigma\,(W_f\ [h_{t-1}, x_t] + b_f) \qquad (3)$$

Where, W_f, b_f is the parameter to be learned in the network, representing weight and offset respectively, h_{t-1} is the output from the previous unit state, x_t is the output of the current unit state, σ Represents the sigmoid function, with a value between 0-1. Its size determines the degree of information reservation in the previous cell. 0 means all information is discarded, and 1 means all information is reserved.

Step 2: Determine the updated information

This step is to decide what new information to store in the cell state, which is mainly completed by the memory gate. There are two main parts here. First, the sigmoid layer determines the extent to which candidate information is updated for the input layer. Second, a tanh layer creates a new candidate information. The calculation formula is shown in Eq. (4) and Eq. (5).

$$i_t = \sigma\,(W_i\ [h_{t-1}, x_t] + b_i) \tag{4}$$

$$\tilde{C}_t = \tanh\,(W_C\ [h_{t-1}, x_t] + b_C) \tag{5}$$

Where W_i, W_C, b_i, b_C is the weight and offset of input layer and tanh layer respectively, i_t represents the reserved information of this node, $\tilde{C}_t$ represents all the new information of this node compared with the previous node.

Step3: Update cell status

This step is to update the cell state based on the calculation results of the previous two gate structures, as shown in Formula (6):

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \tag{6}$$

Where $\tilde{C}_t$ is the updated cell state after calculation in this node.

Step4: Output information

The third step has updated the cell status, which is based on the third step to calculate the output results. The output results are determined by O_t function and a tan layer, O_t determines the degree of information output, and the tanh layer is the activation mapping of the cell state. The expression is shown in Eq. (7) and Eq. (8).

$$O_t = \sigma\,(W_O\ [h_{t-1}, x_t] + b_O) \tag{7}$$

$$\mathrm{h}_t = O_t * \tanh\,(C_t) \tag{8}$$

Where h_t is the output information of this node.

3.4 ADF test and Granger causality test

Granger Causality Test usually acts on stationary series[6,7], so it is necessary to conduct the Augmented Dickey Fuller test on the series before conducting a formal test to determine whether the sequence is stable and prevent the occurrence of pseudo regression.

3.4.1 ADF inspection

When autoregressive modeling is carried out for data, the regression expression with one order lag can be expressed as:

$$\mathrm{y}_t = \rho \mathrm{y}_{t-1} + \mathrm{a} + \boldsymbol{\varepsilon}_t \qquad \mathrm{t} = 1,2\ldots \tag{9}$$

Where a is the constant offset term, $\boldsymbol{\varepsilon}_t$ is a random error, ρ is the coefficient of lag term, and the ADF test is |ρ| Value range of, if |ρ|= 1. Equation (9) becomes a random walk sequence, which is the simplest unit root process. When| ρ|< 1, $\{\mathrm{y}_t\}$ is a stationary sequence. And when| ρ|> 1, we think that $\{\mathrm{y}_t\}$ is a non-stationary sequence with explosive process. Rarely in practice. Before the Granger causality test, the data must be subject to ADF test to judge |ρ| The size of the value, if |ρ|< 1. Granger causality test can be conducted directly if |ρ| = 1. You need to perform differential operation on the data first.

3.4.2 Granger causality test

Granger Causality Test is a powerful relationship testing tool, which was proposed by economist Granger. Granger Causality Test believes that if the prediction effect of using the historical information of X and Y series to predict the future value of Y is more accurate than that of only relying on the historical information of Y to predict the effect of Y, then variable X can be considered as the Granger cause of variable Y. Formula (10) and Formula (11) can be used to explain the details:

$$Y_t = \sum_{i=1}^{s} \alpha_i Y_{t-i} + \mu_{1t} \tag{10}$$

$$Y_t = \sum_{i=1}^{s} \alpha_i Y_{t-i} + \sum_{i=1}^{m} \beta_i X_{t-i} \mu_{2t} \quad (11)$$

If the prediction results obtained by regression with equation (10) significantly improve the explanatory power of the model compared with the results obtained by regression with equation (11), it can be considered that X Granger causes Y, that is, X is the Granger cause of Y. In particular, in the Granger hypothesis test in this paper, as long as one party of X or Y causes the other party, it will be introduced into the model. The original hypothesis of the test is that X is not the Granger cause of Y. The test statistic F can be expressed as:

$$F = \frac{(SSR_R - SSR_u)/m}{SSR_u/[n-(s+m)]} \quad (12)$$

Where m is the number of constraints; S + m is the number of regression coefficients of equation (2); N is the sample size. If the original hypothesis is true, then test the statistics F~F (m, n - (s + m)). If at the significance level α (α=0.05 in this paper), $F>F_\alpha(m, n-(s+m))$, then reject the original hypothesis and think that X Granger causes Y. In addition, the selection of lag order is very important to the results of Granger test. In this paper, the lag order is selected by BIC information criterion.

3.4.3 Result integration

The integration result can be expressed by formula (13)

$$\hat{y}_t = f(\hat{d}_t(1), \hat{d}_t(2), \ldots \hat{d}_t(k)) \quad (k<n) \quad (13)$$

N is the number of modes obtained after the original sequence is decomposed by EEMD, and k is the number of reconstructed modes after DTW reconstruction rules. $\hat{y}_t$ represents the final prediction result of the model in period t, $\hat{d}_t(k)$ represents the prediction results of the model in the k-th reconstructed mode in the t-period. f is the algorithm for the prediction results of the k-th reconstructed mode in the t-period. The final prediction results can be obtained through the f-operation[8.9.10]. Because the original sequence is decomposed into n modes by EEMD, and because of the completeness of the EEMD algorithm, the addition of these n modes must result in the original sequence, and the reconstructed modes are only the addition of some modes, which does not change the decomposition result in essence. Therefore, in this paper, the f algorithm in equation (13) uses simple addition to integrate the results. The result is shown in equation (14):

$$\hat{y}_t = \sum_{i=1}^{k} \hat{d}_t(i) \quad (14)$$

4. Experimental Design

4.1 Experimental data

This paper uses the weekly price data of ethylene glycol from January 6, 2006 to March 19, 2021 as the research sample, with a total of 751 observations. In order to avoid the accidental effect of the model, this paper also analyzes the weekly price data of PX (p-xylene) in the same period (752 observations in total). Among them, the former 80% observation values are used as training sets for model training, and the latter 20% data are used as test sets to test the model effect. The price data of ethylene glycol and PX (p-xylene) are shown in Figure 2 and Figure 3:

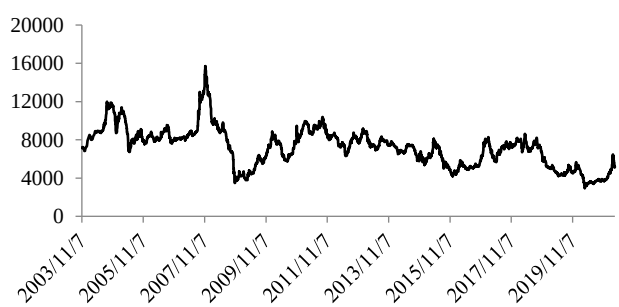

Figure 2: Ethylene glycol price (unit: yuan/ton).

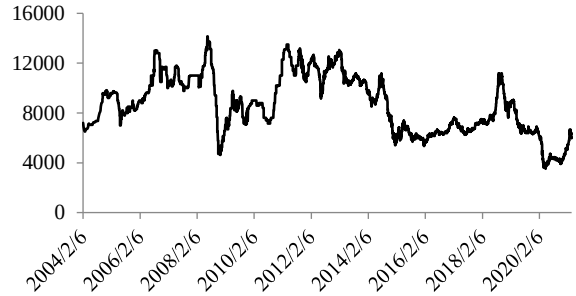

Figure 3: Price of paraxylene (unit: yuan/ton).

4.2 Evaluation indicators of model results

In this paper, two evaluation indicators are selected to measure the prediction effect of the model, namely, the mean percentage absolute error (MAPE) and the root mean square error (RMSE), as shown in Eq. (15) and Eq. (16):

$$\text{MAPE} = \frac{1}{N}\sum_{t=1}^{N}\left|\frac{y_t-\hat{y}_t}{y_t}\right| \tag{15}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(\hat{y}_t-y_t)} \tag{16}$$

Where N is number of samples, $\hat{y}_t$ is the predicted value of t period, y_t is the true value of t period.

In addition, on the basis of MAPE and RMSE, D_{stat} statistical index is introduced to measure the prediction direction accuracy of the model, D_{stat} can be expressed as:

$$D_{stat} = \frac{1}{N}\sum_{t=1}^{N} a_t \tag{17}$$

Where

$$a_t = \begin{cases} 1, & (\hat{y}_{t+1}-y_t)(y_{t+1}-y_t) \geq 0 \\ 0, & (\hat{y}_{t+1}-y_t)(y_{t+1}-y_t) < 0 \end{cases} \tag{18}$$

Finally, this paper also proves the superiority of this model over the benchmark model from the perspective of statistics, using Diebold Mariano (DM) test. The original hypothesis of this test is that there is no significant difference between the prediction effects of model A and model B. Build statistic expression:

$$\text{DM} = \frac{\bar{g}}{\sqrt{(\hat{V}_{\bar{g}}/N)}} \tag{19}$$

Where $\bar{g}$ = 1/N $\sum_{t=1}^{N}((y_t-\hat{y}_{A,t})^2-(y_t-\hat{y}_{B,t})^2)$, $\hat{V}_{\bar{g}}$ =γ_0 +2 $\sum_{j=1}^{\infty}(cov(g_t,g_{t-1}))$, $\hat{y}_{A,t}$, $\hat{y}_{B,t}$ represent the predicted values of model A and model B in period t. DM ~ t (N): perform unilateral t-test on DM statistics. When p < α (α=0.05 in this paper) and DM < 0, reject the null hypothesis and consider that model A is superior to model B.

5. Benchmark model

In order to further demonstrate the effectiveness of the model proposed in this chapter, we also designed a series of benchmark models related to the model in this chapter for comparison on the basis of evaluation indicators. The benchmark model can be summarized as a single model, a composite model without reconstruction process or based on other reconstruction processes, and a composite model based on EMD decomposition process.

The model used in this paper can be expressed as a composite model of EEMD-DTW-LSTM, and the benchmark model mainly includes EMD-DTW-LSTM model based on EMD decomposition; EEMD-NR-LSTM model based on no reconstruction (NR); EEMD-DCD-LSTM model based on data feature driven reconstruction (DCD); EEMD-DTW-ANN model based on feedforward artificial neural network (ANN) and LSTM single model.

In the above model, the integration times of EEMD are uniformly set to 500, and the white noise amplitude is 0.05. ANN is a feedforward neural network with three layers and ReLU activation function. The LSTM step size is set to 8, the number of layers is set to 4, and 10000 times of unified training are performed.

5.1 Empirical analysis

In order to verify the effectiveness of the model framework, the empirical analysis link will use the methods described above to filter the influencing factors, and use the EEMD and DTW reconstruction rules to decompose and reconstruct the price series. Finally, factor sequence and reconstruction sequence are substituted into LSTM network model for time series prediction, and the final prediction result is obtained through simple summation integration method. Based on this, the prediction effect of the model used in this paper is compared with that of the benchmark model. The final comparison results are shown in Figure 3 - Figure 8, Table 2 and Table 3.

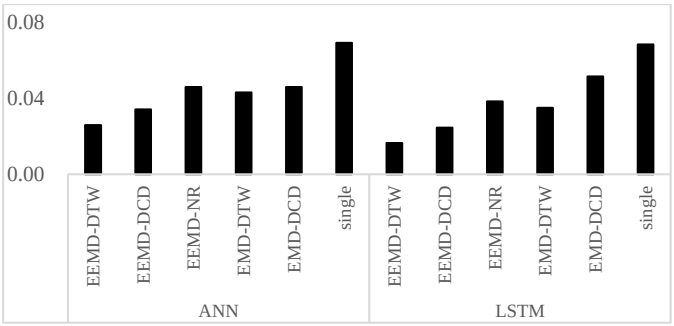

Figure 4: Glycol MAPE.

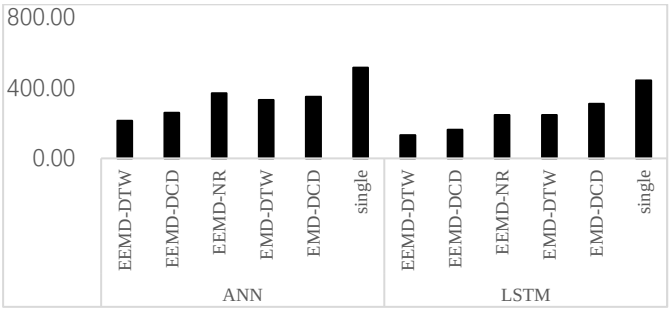

Figure 5: Glycol RMSE.

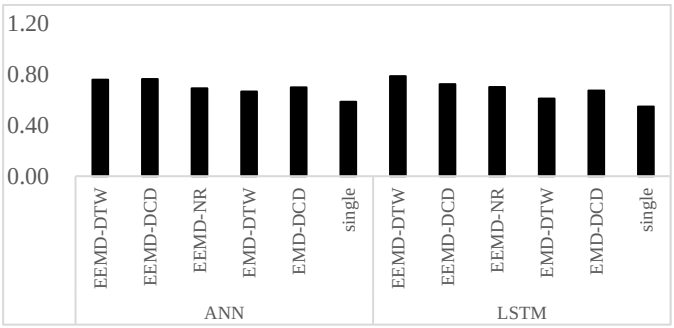

Figure 6: glycol D_{stat}.

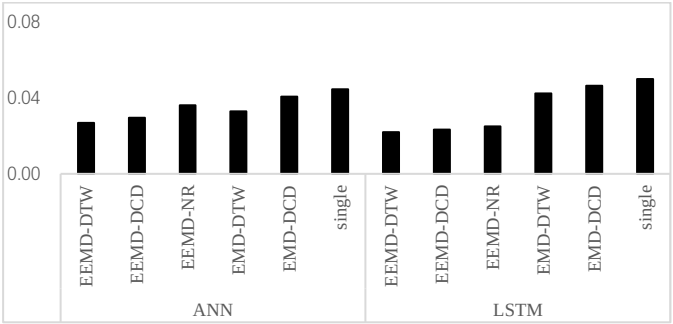

Figure 7: Paraxylene MAPE.

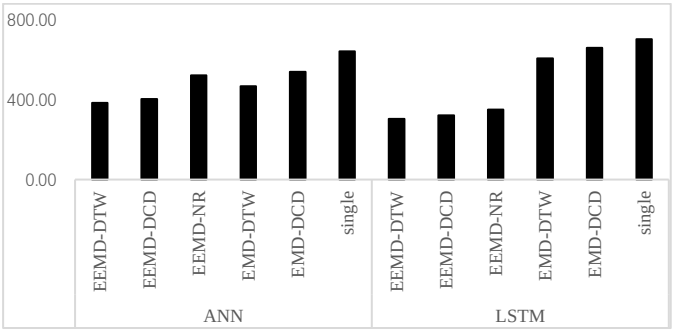

Figure 8: Paraxylene RMSE.

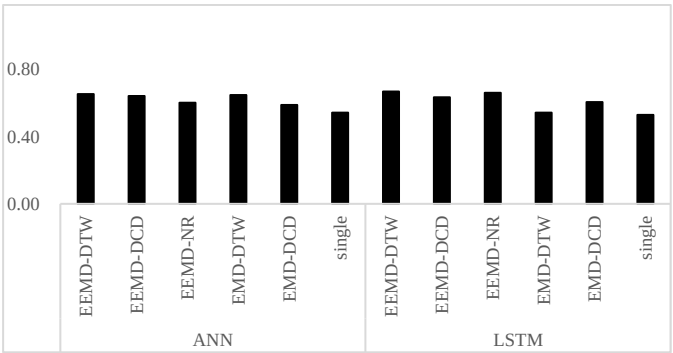

Figure 9: Paraxylene D_{stat}.

Table 2: DM Test Results of LSTM Combination Model for Predicting Ethylene Glycol Price.

	EEMD_ DCD	EEMD_ DTW	EMD_ DTW	EEMD_ NR	single
DCD _ EMD	DM>0 p_valUe <0.05	DM>0 p_value <0.05	DM>0 p_val ue <0.05	DM>0 p_val ue <0.05	DM<0 p_value <0.05
EEMD _ DCD		DM>0 p_value <0.05	DM<0 p_value >0.05	DM<0 p_value >0.05	DM<0 p_value <0.05
EEMD _ DTW			DM<0 p_value <0.05	DM<0 p_value <0.05	DM<0 p_value <0.05
EMD _ DTW				DM<0 p_value >0.05	DM<0 p_value <0.05
EEMD _ NR					DM<0 p_value <0.05

Table 3: DM test results of LSTM combination model for predicting p-xylene price.

	EEMD_ DCD	EEMD_ DTW	EMD_ DTW	EEMD_ NR	single
DCD_ EMD	DM>0 p_value<0.05	DM>0 p_value<0.05	DM>0 p_value<0.05	DM>0 p_value<0.05	DM>0 p_value >0.05
EEMD_ DCD		DM<0 p_value>0.05	DM<0 p_value>0.05	DM>0 p_val ue >0.05	DM<0 p_value <0.05
EEMD_ DTW			DM<0 p_value <0.05	DM<0 p_value <0.05	DM<0 p_value <0.05
EMD_ DTW				DM>0 p_value <0.05	DM<0 p_value <0.05
EEMD_ NR					DM<0 p_value <0.05

From the above results, it can be clearly seen that the LSTM network model based on EEMD decomposition and DTW reconstruction proposed in this chapter has the lowest prediction effect among multiple benchmark models, no matter from the perspective of MAPE or RMSE. And D_{stat} value is obviously higher than that of other benchmark models, which is enough to show that EEMD-DTW-LSTM model has made certain achievements in prediction accuracy and direction, and is superior to its benchmark model. In addition, we also made a comparative analysis from the perspective of statistics. According to the DM test results in Table 2 and Table 3, our models are almost superior to other models at 95% probability.

First, we compare different reconstruction algorithms. Combined with the forecast results of various indicators of the two price-series in Figure 4 - Figure 9, it is not difficult to see that the reconstructed model is obviously better than the model that has not been reconstructed. In addition, the forecast results of the model based on DTW reconstruction are slightly better than those based on DCD reconstruction in most cases, and in a few cases, the DCD reconstruction method will be in the prediction direction indicator D_{stat} value is higher than the DTW reconstruction rule, but compared with the computational complexity of the DCD algorithm, the DTW reconstruction rule still has outstanding advantages.

Thirdly, we compare different decomposition algorithms EMD and EEMD. From the results, no matter what kind of basic monomer model is combined with EEMD decomposition algorithm, the prediction accuracy will be higher than that of EMD algorithm. The main reason is that the EMD algorithm has serious mode aliasing, which makes the original sequence data decomposition not thorough enough, resulting in difficulties in predicting the decomposed modes and irregular problems. The EEMD adopts the set algorithm after adding white noise for many times, which effectively avoids the situation of EMD mode aliasing, and is an effective algorithm more suitable for data decomposition.

Finally, the prediction results of different neural networks are compared and analyzed. The network used in this chapter is LSTM neural network, while the benchmark model uses the ordinary ANN network architecture. From the results, LSTM almost completely outperforms the ordinary ANN network in terms of prediction accuracy and prediction direction. This can be analyzed from two aspects. First, data characteristics. The ANN network is not always inferior to the LSTM network in any case. In this chapter, data characteristics play an important role. In some extreme cases.

We know that the later numerical changes of the price series will be affected to some extent by the previous changes. It can be summarized that the earlier trend will have an effect on the later trend. The data information has a complex temporal correlation with each other, and it is this correlation that gives full play to the advantages of LSTM. Second, data volume. Due to the availability of data, the amount of data used in this paper is not huge, and LSTM's significant advantage is to capture the linkage between small amounts of data. For the ANN network, the lack of data killed the unique advantage of ANN. Combined with the above two points, ANN shows a relatively poor effect in the model.

6. Conclusion

This paper mainly uses the ideas of "decomposing input and combining output" and "proximity principle" to decompose the time series into EEMD and reconstruct DTW. Combining economic factors, the unit root test and Granger causality

test are performed on the corresponding factor series to screen out the main driving factors of the price series, so as to bring them into the model to prepare for the price forecast of multiple factors, Finally, LSTM neural network is used to integrate the prediction results of each reconstructed mode to obtain the final prediction results. And the empirical analysis proves that its prediction effect is better than all kinds of benchmark models.

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